CS 210: Discrete Mathematics *Dr. Mudassir Shabbir, September 3, 2026*

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Worksheet 3

Lecture 3: conditionals and logical equivalence

Exercise 1: Arrow inspection: accept or repair

Let e mean "you enter," s mean "the QR code scans," p mean "the password is correct," and a mean "the app opens." For each proposed translation, write **ACCEPT** if it is correct. Otherwise, write **REPAIR** and give the corrected formula.

1. "You enter only if the QR code scans." Proposed: $s \to e$.

   _____

2. "You enter if the QR code scans." Proposed: $s \to e$.

   _____

3. "A correct password is necessary for the app to open." Proposed: $p \to a$.

   _____

4. "A correct password is sufficient for the app to open." Proposed: $p \to a$.

   _____

Exercise 2: Build, then test, the penguin family

Start with the statement: "If an animal is a penguin, then it is a bird." Write its **converse**, **inverse**, and **contrapositive**. For each one, decide whether it is true. If it is false, give a counterexample.

1. Converse:

   _____

2. Inverse:

   _____

3. Contrapositive:

   _____

Exercise 3: Equivalent — or merely similar?

For each pair, decide whether the two formulas are **logically equivalent**.
Justify your answer with a valid rewrite or one truth assignment that makes the formulas differ.

1. $p \to q$ versus $q \to p$

2. $p \to q$ versus $\neg p \lor q$

3. $(p \land q) \to r$ versus $p \to (q \to r)$

4. $p \to (q \lor r)$ versus $(p \to q) \lor (p \to r)$

Exercise 4: What survives contact with always?

Let t be a tautology and c be a contradiction. Simplify each expression and briefly explain which law you used.

1. $p \land t$

2. $p \land c$